

Group Theory

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Outline

- 1 Flashy flash back
- 2 Homomorphism
- 3 Isomorphism
- 4 Structure of Finite Abelian groups

Flashy flash back

Recap

Previously in part 1, we studied about **Group**, **Subgroup** and its variations like **Cyclic Group**, **Cosets** and **Quotients**.

Notably that we want you to remember the 4 Axioms of Group, how to form a subgroup, and especially, the concept of the "order" of elements inside a **Group** for the upcoming Iso-Homo sections.

Recap Group

A **Group** $(G, \circ)$ satisfies 4 core axioms:

- Closure: $a \circ b = c \in G \ \forall \ a, b \in G$
- Associativity: $(a \circ b) \circ c = a \circ (b \circ c)$ for $a, b, c \in G$
- Identity: $\exists \ e \in G \rightarrow a \circ e = e \circ a = a \ \forall \ a \in G$
- Inverse: $\exists \ a^{-1} \in G$ so that $a \circ a^{-1} = e = a^{-1} \circ a \in G$

For a subset $H \subseteq G$ to be a valid **Subgroup** $(H \leq G)$, we only need to verify two conditions:

- **Closure:** $\forall a, b \in H \implies a \circ b \in H$
- **Inverses:** $\forall a \in H \implies a^{-1} \in H$

Recap Cyclic Groups

- A group G is **Cyclic** if it can be completely generated by a single element $g \in G$:

$$G = \langle g \rangle = \{g^n \mid n \in \mathbb{Z}\}$$

- The **Order of an element** g , denoted $\text{ord}(g)$ or $|g|$, is the smallest positive integer n such that:

$$g^n = e$$

If no such n exists, g has an infinite order (such as $1 \in \mathbb{Z}$).

Recap Cosets and Quotient Groups

Given a subgroup $N \leq G$ and an element $g \in G$:

- A **Left Coset** of N is the shifted set $gN = \{gn \mid n \in N\}$.
- If N is a **Normal Subgroup** ($gNg^{-1} = N \forall g \in G$), the set of all cosets forms a well-defined group under coset multiplication:

$$(aN)(bN) = (ab)N$$

- This group of cosets is the **Quotient Group**, denoted G/N .

Inheritance Properties:

- If G is Abelian $\implies G/N$ is Abelian.
- If G is Cyclic $\implies G/N$ is Cyclic.

Homomorphism

Homomorphism

We learnt about Group and its properties.

But given 2 groups, how do we compare or try to mapping from one to other?

→ Homomorphism

Homomorphism

A homomorphism between groups $(G, \star)$ and $(H, \circ)$ is a map $F : G \rightarrow H$ such that

$$F(g_1 \star g_2) = F(g_1) \circ F(g_2)$$

Example

Compare group $(\mathbb{Z}, +) = (\{Even\} \cup \{Odd\}, +)$ and group $(\mathbb{Z}_2, +)$.

We notice that:

$$even + even = even \qquad 0 + 0 = 0 \pmod{2}$$

$$even + odd = odd \qquad 0 + 1 = 1 \pmod{2}$$

$$odd + even = odd \qquad 1 + 0 = 1 \pmod{2}$$

$$odd + odd = even \qquad 1 + 1 = 0 \pmod{2}$$

If we replace *even* with 0 and *odd* with 1. Their behaviors are exactly the same.

+	even	odd
even	even	odd
odd	odd	even

Table: Cayley Table for $\mathbb{Z}$

+	0	1
0	0	1
1	1	0

Table: Cayley Table for $\mathbb{Z}_2$

Example

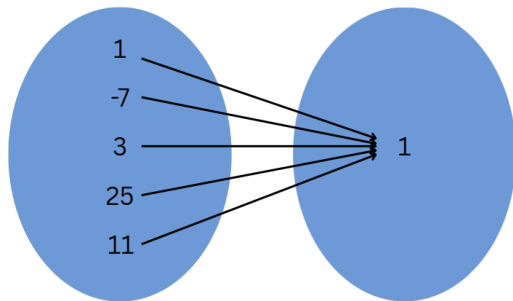
We define a map $F : \mathbb{Z} \rightarrow \mathbb{Z}_2$ with

$$\text{even} \mapsto 0$$

$$\text{odd} \mapsto 1$$

Question: Can we reverse this process?

Example



Since mapping is a well-defined function, we can't just map from 1 to many odd value.

Example

Question: The last 2 groups was easy, given 2 **random** groups.

How do we find a **homomorphism** between them? Or at least there *exists* homomorphism?

And even if we did find one, how can we **reverse** the process?

Example

To answer those questions, we first need to know if a homomorphism does "lose" some information about the group G while mapping to H .

So how do we **measure** the "loss" of information?

The **Kernel** of F , denoted $\ker(F)$, is the set of all elements in G that map to the identity in H .

$$\ker(F) = \{g \in G \mid F(g) = e_H\}$$

Example

Recall our map $F : \mathbb{Z} \rightarrow \mathbb{Z}_2$ where:

- $F(\text{even}) = 0$
- $F(\text{odd}) = 1$

The identity in $\mathbb{Z}_2$ is 0. So, the **Kernel** is the set of all elements that map to 0:

$$\ker(F) = \{\dots, -4, -2, 0, 2, 4, \dots\} = 2\mathbb{Z}$$

What did we lose? We lost the "size" of the numbers. Both 2 and 2,000,000 are now just "0." The kernel "collapsed" the infinite integers into a single point.

The **Image** of F , denoted $\text{im}(F)$, is the set of all elements in H that are "hit" by the mapping.

$$\text{im}(F) = \{F(g) \mid g \in G\}$$

Example

In the same map $F : \mathbb{Z} \rightarrow \mathbb{Z}_2$:

$$\text{im}(F) = \{F(n) \mid n \in \mathbb{Z}\}$$

Since every integer is either even or odd, our outputs are only $\{0, 1\}$.

$$\text{im}(F) = \{0, 1\} = \mathbb{Z}_2$$

This map is **surjective** (onto) because it contains every element in H .

Question: Can $\text{im}(F)$ cover more/less elements than H ?

Example

If $|\text{im}(F)| < |H|$, the homomorphism is **not surjective**.

This happens when:

- **Size Mismatch:** G is too "small" or simple to cover H .
- **Structure Mismatch:** H doesn't have elements that "fit" the rules of G .

Example: Map $F : \mathbb{Z}_2 \rightarrow \mathbb{Z}_{10}$ by $F(1) = 5$.

The image is only $\{0, 5\}$. Even though $\mathbb{Z}_{10}$ has 10 elements, the image only has 2. It is a valid homomorphism, but we "missed" most of H .

If $F : G \rightarrow H$ is a homomorphism, then:

- **Identity:** $F(e_G) = e_H$.
- **Inverses:** $F(g^{-1}) = [F(g)]^{-1}$.
- **Orders:** $\text{ord}(F(g))$ divides $\text{ord}(g)$.
- **Injectivity:** F is injective $\iff \ker(F) = \{e_G\}$.

Proposition 1: $F(e_G) = e_H$.

Suppose that e_G and e_H are identities of G and H . We have:

$$e_H F(e_G) = F(e_G) = F(e_G e_G) = F(e_G) F(e_G)$$

By **Cancellation** rule, $e_H = F(e_G)$.

Proposition 2: $F(g^{-1}) = [F(g)]^{-1}$.

This proposition follow from the first one:

$$e_H = F(e_G) = F(gg^{-1}) = F(g)F(g^{-1}).$$

Hence:

$$e_H = F(g)F(g^{-1}) \iff [F(g)]^{-1} = F(g^{-1})$$

Properties

Proposition 3: $\text{ord}(F(g))$ divides $\text{ord}(g)$.

This property is easy to see, as a Homomorphism $F : G \rightarrow H$ must follow:

$$F(g_1 \circ g_2) = F(g_1) \star F(g_2) \quad \forall g_1, g_2 \in G$$

Let n be the order of $g \in G$ (i.e., $n = \text{ord}(g)$). Then:

$$e_H = F(e_G) = F(\underbrace{g \circ g \cdots \circ g}_{n \text{ times}}) = \underbrace{F(g) \star F(g) \star \cdots \star F(g)}_{n \text{ times}} = F(g)^n$$

Which means that $F(g)^n = e_H$ holds true. But knowing this does not mean $\text{ord}(F(g)) = n$, **but rather that n is a multiple of $\text{ord}(F(g))$.**

Therefore, $\text{ord}(F(g))$ divides $\text{ord}(g)$.

Proposition 4: F is injective $\iff \ker(F) = \{e_G\}$.

This proposition follow from the first one too. If F is injective and $F(e_G) = e_H$ then:

$$\ker(F) = \{g \in G \mid F(g) = e_H\} = \{e_G\}$$

Back to the Question

Earlier, we asked: *Given two random groups, how do we find a homomorphism between them? Or at least there exists homomorphism?*
And for the latter question, the answer is yes.

For any groups G and H , there is always at least one homomorphism: the **trivial homomorphism**

$$F(g) = e_H \quad \forall g \in G.$$

Since

$$F(ab) = e_H = e_H e_H = F(a)F(b),$$

this map is always a valid homomorphism.

Now the existence question is easy, the more interesting question is:

How do we construct a non-trivial homomorphism?

Back to the Question

The answer lies in the **Orders** property:

$$\text{ord}(F(g)) \mid \text{ord}(g).$$

Instead of blindly mapping every element, we use this constraint to divide our approach into **two structural cases**:

- **Case 1:** Group G is **Cyclic**.
- **Case 2:** Group G is **Not Cyclic**.

Case 1: How to find a Homomorphism

Since a homomorphism preserves the group operation, for a **cyclic group** $G = \langle g \rangle$, it suffices to choose the image of the generator g .

The order condition

$$\text{ord}(F(g)) \mid \text{ord}(g)$$

restricts the possible choices for $F(g)$. Once $F(g)$ is chosen, every other element is determined by

$$F(g^n) = F(g)^n.$$

Case 1: Example

For example: Constructing a map $F : \mathbb{Z}_{15} \rightarrow \mathbb{Z}_{18}$ (under $+$)

- 1 $\mathbb{Z}_{15}$ is cyclic, generated by 1.
- 2 A homomorphism is completely determined by where 1 is sent.
- 3 Since 1 has order 15, its image must have order dividing 15.
- 4 In $\mathbb{Z}_{18}$, the elements satisfying

$$15a \equiv 0 \pmod{18}$$

are $a = 0, 6, 12$.

- 5 Choose $F(1) = 6$. Then

$$F(x) = 6x \pmod{18}.$$

The image is

$$\text{im}(F) = \{0, 6, 12\},$$

which is a proper subgroup of $\mathbb{Z}_{18}$. The homomorphism is valid even though it does not cover all of $\mathbb{Z}_{18}$.

Case 2: How to find a Homomorphism

Although G is not cyclic, we can use a similar idea as in Case 1.

Instead of a single generator, choose a generating set

$$G = \langle g_1, g_2, \dots, g_k \rangle.$$

We define the images of these generators and then extend the map to all elements of G by preserving the group operation.

The main challenge is ensuring that the chosen images satisfy all relations among the generators.

Case 2: Example

Example: Constructing a map $F : \mathbb{Z}_2 \times \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$

- 1 **Find the Generating Set:** Since $\mathbb{Z}_2 \times \mathbb{Z}_2$ is not cyclic, we need a team of generators: $\langle (1, 0), (0, 1) \rangle$.
- 2 **Apply the Order Filter:** Both generators have an order of 2. Their target images in $\mathbb{Z}_2$ must have an order dividing 2.
- 3 **Map the Generators:** Let's choose the following targets:

$$F(1, 0) = 1, \quad F(0, 1) = 0$$

- 4 **Extend to the Entire Group:** Because F must preserve the addition operation, any arbitrary element (a, b) maps as:

$$F(a, b) = aF(1, 0) + bF(0, 1) = a(1) + b(0) = a \pmod{2}$$

Our mapping yields: $F(0, 0) = 0$, $F(1, 0) = 1$, $F(0, 1) = 0$, $F(1, 1) = 1$.

The image is $\text{im}(F) = \mathbb{Z}_2$, meaning F is a **surjective** homomorphism.

Homomorphisms and Subgroups

A vital takeaway of homomorphisms is how they preserve sub-structures:

- If S is a subgroup inside G , its target image $F(S)$ will form a perfectly valid subgroup inside H .
- If S' is a subgroup inside H , its pulled-back pre-image $F^{-1}(S')$ will form a valid subgroup inside G .

Note:

The trivial set $\{e_H\}$ is a subgroup of H . Its pre-image is the definition of the Kernel. Therefore, **the Kernel is guaranteed to be a subgroup of G .**

Isomorphism

Isomorphism

Previously, we studied about homomorphism and noticed that we can map $F : \mathbb{Z} \rightarrow \mathbb{Z}_2$ but not the other way.

But in what situation *can* we reverse the mapping? What if the groups were a "perfect match"?

Isomorphism

We can reverse the process only when the mapping is **Bijjective** (1-to-1 and Onto). Two groups G and H are **Isomorphic**, denoted $G \cong H$, if:

- 1 There is a homomorphism $F : G \rightarrow H$.
- 2 Each element $g \in G$ maps to a **unique** element in H , and every element in H is covered.

→ In this case, G and H are essentially the **same group**.

Example

Compare group $(\mathbb{Z}_4, +)$ and group $(\{1, i, -1, -i\}, \times)$.

We notice that:

+	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Table: Cayley Table for $\mathbb{Z}_4$

$\times$	1	i	-1	-i
1	1	i	-1	-i
i	i	-1	-i	1
-1	-1	-i	1	i
-i	-i	1	i	-1

Table: Cayley Table for $\{1, i, -1, -i\}$

Do you see the pattern here?

Example

+	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Table: Cayley Table for $\mathbb{Z}_4$

$\times$	1	i	-1	$-i$
1	1	i	-1	$-i$
i	i	-1	$-i$	1
-1	-1	$-i$	1	i
$-i$	$-i$	1	i	-1

Table: Cayley Table for $\{1, i, -1, -i\}$

Example

The color pattern shows that these two groups are structurally the same. We define the isomorphism $F : \mathbb{Z}_4 \rightarrow \{1, i, -1, -i\}$ as:

$$0 \mapsto 1$$

$$1 \mapsto i$$

$$2 \mapsto -1$$

$$3 \mapsto -i$$

Every addition in $\mathbb{Z}_4$ corresponds to a multiplication in the complex set.

This is a **perfect, reversible map**.

Properties

If $G \cong H$ via the map $F : G \rightarrow H$, then G and H share all structural properties:

- **Order Equality:** $|G| = |H|$.
- **Abelian:** G is abelian if and only if H is abelian.
- **Cyclic:** G is cyclic if and only if H is cyclic.
- **Invertibility:** The inverse map $F^{-1} : H \rightarrow G$ is also an isomorphism.

If any of these fail, the groups cannot be isomorphic.

First Isomorphism Theorem

Let

$$F : G \rightarrow H$$

be a group homomorphism. Then

$$G / \ker(F) \cong \operatorname{im}(F).$$

More precisely, the map

$$\bar{F} : G / \ker(F) \rightarrow \operatorname{im}(F), \quad g \ker(F) \mapsto F(g)$$

is an isomorphism.

Kernel = information lost, Image = information preserved.

Why Is $\overline{F}$ Well-Defined?

The map

$$\overline{F}(g \ker(F)) = F(g)$$

must not depend on which representative g we choose.

Suppose

$$g \ker(F) = g' \ker(F).$$

Then

$$g^{-1}g' \in \ker(F).$$

Therefore,

$$F(g^{-1}g') = e_H.$$

So

$$F(g)^{-1}F(g') = e_H,$$

which implies

$$F(g) = F(g').$$

Thus, $\overline{F}$ is well-defined.

First Isomorphism Theorem: Parity Map

For the parity map

$$F : \mathbb{Z} \rightarrow \mathbb{Z}_2, \quad F(n) = [n]_2,$$

we have

$$\ker(F) = 2\mathbb{Z}$$

and

$$\operatorname{im}(F) = \mathbb{Z}_2.$$

Therefore,

$$\boxed{\mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}_2.}$$

The quotient has two cosets:

$$2\mathbb{Z} \quad \text{and} \quad 1 + 2\mathbb{Z}.$$

Counting Formula for Finite Groups

If G is finite, the First Isomorphism Theorem gives

$$G / \ker(F) \cong \operatorname{im}(F).$$

Therefore,

$$|\operatorname{im}(F)| = |G / \ker(F)| = \frac{|G|}{|\ker(F)|}.$$

Equivalently,

$$|G| = |\ker(F)| |\operatorname{im}(F)|.$$

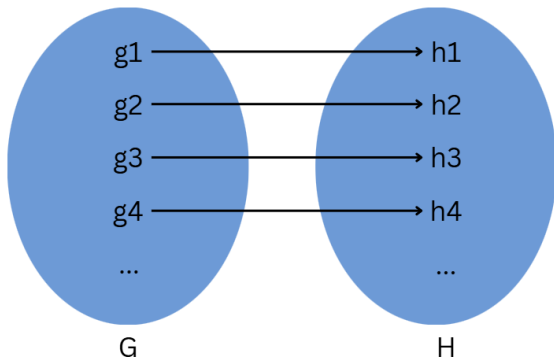
This formula helps us compute the size of the image without listing every output.

Properties

Proposition 1: $|G| = |H|$.

Proof:

This is self-explanatory since $G \cong H$, that mean the mapping is *Bijjective*.



→ Preserving group structure (including cardinality).

Proposition 2: G is abelian if and only if H is abelian.

Proof:

Suppose that h_1 and h_2 are elements of H . Since F is onto, there exist elements g_1 and g_2 such that $F(g_1) = h_1$ and $F(g_2) = h_2$. Therefore:

$$h_1 h_2 = F(g_1)F(g_2) = F(g_1 g_2) = F(g_2 g_1) = F(g_2)F(g_1) = h_2 h_1.$$

$\rightarrow H$ is abelian

Proposition 3: G is cyclic if and only if H is cyclic.

Proof:

Suppose that g is generator of group $G = \langle g \rangle$. We want to show that group H is also **generated** by $F(g)$.

Since F is onto, there must be an element $x \in G$ such that $h = F(x)$. Because of $G = \langle g \rangle$, $x = g^n$. Hence:

$$h = F(g^n) = \underbrace{F(g)F(g)\dots F(g)}_{n \text{ times}} = F(g)^n$$

$\rightarrow H = \langle F(g) \rangle$ is cyclic

Direct product

Given two groups G and H , it is possible to construct a new group from the Cartesian product of G and H , $G \times H$. Conversely, given a large group, it is sometimes possible to **decompose** the group.

That is, a group is sometimes isomorphic to the **direct product** of two smaller groups.

Rather than studying a large group G , it is often easier to study the component groups of G .

Direct product

Given groups $(G, \circ)$ and $(H, \bullet)$, define the set

$$G \times H = \{(g, h) : g \in G, h \in H\}$$

with operation defined componentwise:

$$(g_1, h_1) * (g_2, h_2) := (g_1 \circ g_2, h_1 \bullet h_2).$$

If we use additive notation in both groups:

$$(g_1, h_1) + (g_2, h_2) = (g_1 + g_2, h_1 + h_2).$$

Direct product

Let $(G, \star)$ and $(H, \circ)$ be groups.

The set $G \times H$ is a **group** under the operation:

$$(g_1, h_1)(g_2, h_2) = (g_1 \star g_2, h_1 \circ h_2)$$

Where $g_1, g_2 \in G$ and $h_1, h_2 \in H$.

For the sake of simplicity, we'll write (g_1g_2, h_1h_2) .

Clearly the binary operation defined above is closed ($g_1g_2 \in G, h_1h_2 \in H$).

If e_G and e_H are the identities of the groups G and H respectively, then (e_G, e_H) is the identity of $G \times H$.

The inverse of $(g, h) \in G \times H$ is (g^{-1}, h^{-1}) . The fact that the operation is associative follows directly from the associativity of G and H .

Example

Let R be the group of real numbers under addition.

The Cartesian product of R with itself, $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$, is also a group, in which the group operation is just addition in each coordinate.

That is, $(a, b) + (c, d) = (a + c, b + d)$. The identity is $(0, 0)$ and the inverse of (a, b) is $(-a, -b)$.

What about Direct Product of many **groups**? Say a direct product

$$\prod_{i=1}^n G_i = G_1 \times G_2 \times \cdots \times G_n$$

of the groups $G_1, G_2, \dots, G_n$ is defined in exactly the same manner. If $G = G_1 = G_2 = \cdots = G_n$, we often write G^n instead of $G_1 \times G_2 \times \cdots \times G_n$.

Example

The group $\mathbb{Z}_2^n = \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \dots$, considered as a set, is just the set of all binary n -tuples.

The group operation is the “exclusive or” of two binary n -tuples. For example:

$$(01011101) + (01001011) = (00010110)$$

Proposition 1:

Let $(g, h) \in G \times H$. If g and h have finite orders r and s respectively, then the order of (g, h) in $G \times H$ is the least common multiple of r and s .

Proposition 2:

The group $\mathbb{Z}_m \times \mathbb{Z}_n$ is isomorphic to $\mathbb{Z}_{mn}$ if and only if $\gcd(m, n) = 1$.

Proposition 1:

Suppose that $m = \text{lcm}(r, s)$ is the least common multiple of r and s and let $n = |(g, h)|$. Then

$$(g, h)^m = (g^m, h^m) = (e_G, e_H)$$

$$(g^n, h^n) = (g, h)^n = (e_G, e_H)$$

Hence, n must divide m , and $n \leq m$.

However, by the second equation, both r and s must divide n ; therefore, n is a common multiple of r and s .

Since m is the least common multiple of r and s , $m \leq n$. Consequently, m must be equal to n .

Proposition 2:

We will first show that if $\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn}$, then $\gcd(m, n) = 1$.

We will prove the contrapositive. That is, we will show that if $\gcd(m, n) = d > 1$, then $\mathbb{Z}_m \times \mathbb{Z}_n$ cannot be isomorphic to $\mathbb{Z}_{mn}$.

We'll prove this via proving $\mathbb{Z}_m \times \mathbb{Z}_n$ cannot be cyclic.

Notice that mn/d is divisible by both m and n ; hence, for any element $(a, b) \in \mathbb{Z}_m \times \mathbb{Z}_n$,

$$\underbrace{(a, b) + (a, b) + \cdots + (a, b)}_{mn/d \text{ times}} = (0, 0)$$

Therefore, no (a, b) can generate all of $\mathbb{Z}_m \times \mathbb{Z}_n$.

Proof

This results in $\mathbb{Z}_m \times \mathbb{Z}_n$ cannot be cyclic.

Because of $\mathbb{Z}_{mn}$ is cyclic.

Therefore, $\mathbb{Z}_m \times \mathbb{Z}_n$ cannot be isomorphic to $\mathbb{Z}_{mn}$ if $\gcd(m, n) = d > 1$.

Example

Consider the groups $\mathbb{Z}_{12}$ and $\mathbb{Z}_{60}$.

Let $(8, 56) \in \mathbb{Z}_{12} \times \mathbb{Z}_{60}$.

Since $\gcd(8, 12) = 4$, the order of 8 is $12/4 = 3$ in $\mathbb{Z}_{12}$.

Similarly, the order of 56 in $\mathbb{Z}_{60}$ is 15.

The least common multiple of 3 and 15 is 15; hence, $(8, 56)$ has order 15 in $\mathbb{Z}_{12} \times \mathbb{Z}_{60}$.

Structure of Finite Abelian Groups

Fundamental theorem of finite abelian groups

We next state a theorem that classifies all finite abelian groups up to isomorphism.

Theorem

A finite abelian group (with more than one element) is isomorphic to a direct product of cyclic groups

$$\mathbb{Z}_{p_1^{e_1}} \times \cdots \times \mathbb{Z}_{p_r^{e_r}},$$

where the p_i are primes (not necessarily distinct) and the e_i are positive integers. This direct product of cyclic groups is unique up to the order of the factors.

Fundamental theorem of finite abelian groups

Alternatively:

Theorem

A finite abelian group (with more than one element) is isomorphic to a direct product of cyclic groups

$$\mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_t},$$

where each $m_i > 1$, and where for $i = 1, \dots, t - 1$, we have $m_i \mid m_{i+1}$. Moreover, the integers $m_1, \dots, m_t$ are uniquely determined, and m_t is the exponent of the group.

"The statements of these theorems are much more important than their proofs." said Victor Shoup. I agree

Fundamental theorem of finite abelian groups

Luckily, the latter one :

Let G be a finite abelian group, $G \neq \{0\}$. Then there exist integers

$$1 < m_1 \mid m_2 \mid \cdots \mid m_t$$

such that

$$G \cong \mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_t}.$$

Was proved by Shoup as a lemma. I haven't fully understood the proof, yet can only give you a brief overall look at how it was proven

Fundamental theorem of finite abelian groups

Let $m = \exp(G)$, the least positive integer such that $mx = 0$ for all $x \in G$.

Two facts from earlier in the chapter are crucial:

- ① There exists an element $a \in G$ with

$$\text{ord}(a) = m.$$

- ② If $A, B \leq G$ satisfy

$$A \cap B = \{0\},$$

then

$$A + B \cong A \times B.$$

Now, Let G be a finite abelian group of exponent m , and let $a \in G$ have order m . Set

$$A = \langle a \rangle.$$

Then there exists a subgroup $B \leq G$ such that

$$G = A \oplus B.$$

Once this lemma is proved, induction on $|G|$ gives

$$B \cong \mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_{t-1}},$$

and hence

$$G \cong \mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_{t-1}} \times \mathbb{Z}_m.$$

Since $\exp(B) \mid m$, we obtain the divisibility chain

$$m_1 \mid \cdots \mid m_{t-1} \mid m.$$

This suffering has come to an end. Thank you for listening